

QUANG MINH NGUYEN

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Education

Florida State University, Bachelor of Science in Computer Science (GPA: 3.8) **Expected May 2027**

- **Awards:** Shellhacks 2025 Hackathon Winner, 3x FSU ACM Programming Contest Winner, President's List
- **Relevant Coursework:** Object-oriented Programming (OOP), Data Structures and Algorithms, Operating Systems, Database Management Systems, Software Engineering, Applied Machine Learning.

Experience

Arcanic AI

May 2026 - Present

Software Engineer Intern

Hanoi, Vietnam

- Built an **AI multi-agent** legal reference-and-consultation platform using **LangGraph**, supporting policy lookup, impact analysis, and consistency review over 160,000 legal documents.
- Improved retrieval context recall from 82% to 91% and speed by 40% on 160,000 Vietnamese legal corpus by pairing **GraphRAG knowledge-graph traversal with HNSW-indexed vector search** in PostgreSQL.
- Built an **LLM query-processing** layer using **HyDE, multi-query expansion, and self-reflective refinement** that rewrites and splits complex legal questions into sub-queries, boosting retrieval accuracy on multi-part questions.
- Co-built a **ReBAC** authorization system enforcing **fine-grained, relationship-derived permissions** on sensitive legal cases, granting each user access only through their relationship to a case.

Viet Tech Mentorship Program

April 2025 - Dec 2025

Software Engineer Fellow

Remote (New York, NY)

- Developed Treverse, a full-stack application using **MongoDB, Express, NodeJS, ReactJS** in a team of 10, enabling users to manage and track job applications, share interviews, and search postings efficiently.
- Reduced data retrieval time **60% (8s → 3s)** by implementing cursor-based pagination and Tanstack frontend caching.
- Architected **10+ REST APIs** using **controller-service-repository pattern** to enforce separation of concerns.
- Ensured backend robustness with **40+ unit and integration tests using Mocha, Chai, and Supertest**.
- Engineered an automated interview-resources pipeline using **Gemini API, AWS Lambda, and Redis**.
- Collaborated with product owner and product designer on user research and feature ideation, refining flows with **ShadCN components and TailwindCSS** to improve usability and support long-term user retention.

AWS Cloud Club

Feb 2026 - Present

Technical Lead

Tallahassee, FL

- Leading a 6-member backend team to build an image processing system using **AWS Lambda, S3, DynamoDB**, leveraging **AI-assisted development tools (Cursor)** to accelerate API scaffolding and testing.
- Coordinating with frontend, cloud, and documentation teams to define **APIs, data schemas, and integration**.
- **Researched AWS services and system design**, created technical presentations, and onboarded backend members.

Projects

Recall AI 🌀 | FastAPI, React, Postgres, LangGraph, OpenSearch

March 2026

- Built a RAG-powered note-taking platform with workspace-scoped retrieval using OpenSearch vector search.
- Implemented hybrid search (BM25 + KNN) with reranking to improve retrieval accuracy.
- Developed a LangGraph-based agent system for quiz generation, flashcards, and summarization.

Open Pothole Map 🌀 | 2nd Place – Waymo Challenge, ShellHacks 2025 | Devpost

Sep 2025

- Integrated **Google Map JavaScript API, Geolocation API, and Device Orientation API** to enable live user tracking, real-time pothole mapping, and navigation alerts, enhancing user engagement and app utility.
- Designed a **WebSocket-based** ingestion pipeline to receive real-time camera frames from clients, perform server-side pothole detection, and persist detected images to **Google Cloud Storage**.
- Implemented secure **OAuth 2.0 authentication via Firebase** for user sign-in and session management.

Technical Skills

Languages: JavaScript, TypeScript, Python, Java, C++, SQL, HTML/CSS, Assembly

Technologies/Frameworks: React.js, LangChain/LangGraph, Node.js, FastAPI, Spring Boot, Redis, Docker, Git/Github

Cloud: Amazon Web Services (AWS), Google Cloud Platform (GCP)

Database: MySQL, PostgreSQL, MongoDB

Skills: Web Development (Full-Stack), Frontend, Backend, Data Structures, Algorithms and Problem-Solving